Zhakhangir Mamayev

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Education

Boston University Sep 2023 – Dec 2026
Bachelor of Science in Computer Engineering *GPA: 3.69*
Relevant Coursework Operating Systems, Applied Algorithms, Cybersecurity, Software Engineering

Experience

ReCo Digital Feb 2025 – Present
AI Developer Intern *Sydney, Australia*

- Built and deployed a Flask-based full-stack web app (frontend + REST endpoints) that leverages a multimodal Gemini Pro 2.5 model and OCR to extract itemized receipt data, cutting manual data-entry time by **90%** and achieving **92%** extraction accuracy on unstructured receipts.
- Designed backend data models and migration pipelines using SQLAlchemy + Alembic (Flask-Migrate) and PostgreSQL (Supabase), implemented DB migrations and schema management for production deployments.
- Implemented robust OCR post-processing, Pydantic-style validation patterns for AI outputs, scalable multi-file upload handling, and real-time food-weight estimation
- Retained for a paid follow-on contract to scale OCR pipeline and enable monthly-view data editing and default-weight fallbacks.

Boston University Mars Rover Club Sep 2024 – Dec 2024
Computer Vision Engineer *Boston, MA*

- Implement ROS camera and LiDAR pipelines to simulate rover motion in Gazebo and validate perception algorithms in CI-like simulation scenarios.
- Author Xacro robot models and simulation configs to reduce setup time for teammates and accelerate iteration on perception stacks.
- Lead debugging of ROS migration issues to restore sensor fidelity and keep test coverage for perception modules.

Projects

Muscle Intelligence | *Python, TensorFlow, OpenCV, imageio* Oct 2024

- Built a mobile app at HackHarvard to improve workout form using multi-camera keypoint detection.
- Improved keypoint reading accuracy by **30%** through targeted imageio preprocessing and model tuning.
- Optimized a cosine-similarity pipeline for robust multi-view pose matching.

Boxing Dynamometer | *C++, Arduino, ESP32* Oct 2024

- Engineered a device to measure punch force and present results mechanically; developed a multi-read aggregation algorithm for reliable peak-force detection.
- Extended IMU range from **4g** to **16g** by modifying sensor registers to capture high-impact events.
- Implemented Bluetooth reporting for real-time result visualization.

User-space threads implementation | *C, Linux* Oct 2024

- Implemented user-space POSIX-like threads with custom scheduler and jump-buffer context switching for educational OS project.
- Added signal-driven scheduling hooks to enable pluggable scheduling policies and ease experimentation.

Oqiga.AI | *Python, Tortoise TTS, Google Colab* Feb 2023

- Built a voice-synthesis pipeline to reproduce parental voice characteristics with **81%** subjective similarity in evaluations; placed top 3 at MakeHarvard (Infosys Best Use of AI).
- Used Colab for training/inference and established reproducible notebooks for demonstration.

Technical Skills

Languages: Python, C, C++, JavaScript, MATLAB
Frameworks / Libraries: Flask, TensorFlow, OpenCV, ROS, Gazebo, Pandas, NumPy, Matplotlib
Tools / Platforms: Git, GitHub, Google Colab, PyCharm, VS Code, Arduino, ESP32
Other: REST APIs, OCR pipelines, TTS, unit testing